Sanketh Edara

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Education

Purdue University | Bachelor of Science West Lafayette, IN
Dual Major: *Computer Science and Data Science* ***Minors:*** *Finance, Mathematics* *May 2027*

GPA: 3.84 **Honors:** Dean's List, Semester Honors
Awards: 2x Hackathon Winner **Organizations:** ML@Purdue, PurdueHackers
Relevant Coursework: Systems Programming, Computer Architecture, Data Structures and Algorithms, Analysis of Algorithms, Artificial Intelligence, Data Mining and Machine Learning, Relational Databases, Programming in C, Object-Oriented Programming, Discrete Mathematics

Technical Skills

Systems & Performance: C/C++, Embedded Systems, Linux/Unix, Real-Time Systems
Robotics / Perception: Sensor Fusion, SLAM, Computer Vision, OpenCV, ROS
Languages: C/C++, TypeScript, Python, Java, SQL
Frameworks: React, Node.js, Flask, Django
Tools: Git, Docker, AWS (EC2, S3), GCP
ML / AI: PyTorch, TensorFlow, Scikit-learn, LSTM, ARIMA

Experience

Interim Engineering Intern | C++, Embedded Systems May 2025 – Aug 2025
Qualcomm Technologies, Inc.

- Built a **real-time sensor fusion pipeline** using an Extended Kalman Filter (EKF) to fuse IMU data (accelerometer + gyroscope) for orientation tracking on XR devices.
- Deployed algorithm on **Hexagon DSP (Snapdragon SM8150)** and integrated with Sensor Execution Environment (SEE) for continuous real-time data ingestion.
- Optimized performance using Hexagon Vector Extensions (HVX), reducing computational latency and improving DSP efficiency by **30%**.
- Achieved significant power savings by offloading computation from CPU to DSP (up to **90% reduction** in average power usage).
- Debugged and profiled **low-latency real-time pipelines** in a Linux-based embedded environment.

ML Software Engineering Co-Op | Python, APIs, Data Pipelines Aug 2023 – May 2024
Caterpillar Inc.

- Built **ML forecasting pipelines** (NBEATS, ARIMA, LSTM) exposed via internal APIs for EV fleet prediction systems.
- Designed structured **JSON-based data pipelines** for model inputs/outputs and dashboard integration.
- Prototyped a **retrieval-based analytics assistant** over EV telemetry data to surface operational insights.
- Evaluated model performance using scenario-based validation and error metrics to ensure production reliability.
- Improved fleet efficiency and reduced operational uncertainty through AI-driven forecasting.

Software Engineering Intern | Python, LLM APIs, NLP May 2024 – Aug 2024
QXO

- Built an **LLM-powered speech evaluation system** using Whisper-1 and NLP pipelines to analyze presentation quality.
- Designed retrieval + embedding pipelines (TF-IDF, BERT) to **ground model outputs** and improve scoring accuracy.
- Developed modular pipeline architecture (transcription → feature extraction → scoring → feedback generation).
- Automated evaluation workflows using AI assistants, reducing manual effort by **75%**.
- Prototyped an **agent-style pipeline** coordinating multiple tools for end-to-end evaluation.

Projects

Undergraduate Research – Swarm Robotics (SLAM-Based Disaster Response) | Python, Java, ROS, Lidar

- Built a **real-time SLAM pipeline** using Hector SLAM to map environments from Lidar data.
- Programmed TurtleBots with **ROS** for autonomous navigation and localization.
- Designed decentralized **swarm coordination strategies** for multi-agent systems.
- Evaluated performance using **ANOVA** across simulation scenarios.

AI COO Multi-Agent System (Startup Operations Automation) | Python, FastAPI, Redis

- Built a **multi-agent system** to automate startup operations using LLM-driven agents.
- Designed a **distributed event-driven architecture** with an Event Bus for agent communication.
- Implemented a **global context store + agent memory** for shared state and decision-making.
- Integrated APIs (Gmail, webhooks) and LLM pipelines for end-to-end workflows.